Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to Fe-56

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ABSTRACT

Fast-neutron fluence to the reactor pressure vessel is a key quantity for PWR irradiation and life-extension assessments. Nuclear data often supply a large share of the uncertainty, and the literature has largely emphasized cross-section magnitudes; however, in attenuation-dominated transport, uncertainties in scattering angular distributions (ADs) also matter and are less routinely propagated. We quantify the effect of Fe-56 elastic-scattering AD uncertainties by propagating data from two general-purpose libraries (JEFF-4.0/3.3, JENDL5) using linear perturbation theory and Monte Carlo sampling. Two complementary systems are considered: a spherical PWR-inspired toy model representing core-to-ex-vessel attenuation, and the SINBAD ASPIS-Iron benchmark. Across both systems, the two libraries yield different uncertainty estimates, with JENDL-5 giving larger values in the sphere and comparable ones in ASPIS. When compared to EXFOR differential measurements, JENDL's uncertainties more often cover the discrepancies between evaluations and experiments, in comparison to the data in the JEFF libraries. In both cases, AD-driven contributions are smaller than those from angular-integrated cross sections, yet can be significant depending on the adopted covariance evaluation and should be accounted for RPV-fluence uncertainty assessments. These results highlight the value of continued development and documentation of covariance data for scattering angular distributions to strengthen general-purpose evaluations and PWR irradiation calculations.

Keywords: Fast-neutron fluence, Elastic-scattering angular distribution, Nuclear-data covariances, Fe-56, Uncertainty quantification

1. INTRODUCTION

Reactor pressure vessel (RPV) integrity assessments for life-extension of pressurized water reactors (PWRs) rely on accurate knowledge of fast neutron fluence and its uncertainty [1]. Neutron irradiation causes embrittlement of the RPV steel, so fluence estimates with quantified uncertainties are needed to ensure safety margins for long-term operation. In the dosimetry calculations used to determine RPV fluence, the nuclear data uncertainties can be non-negligible, where traditionally, most attention has focused on cross-section uncertainties [2, 3]. However, in attenuation-dominated shielding scenarios like core-to-vessel neutron transport, the angular distribution of scattering can significantly influence how neutrons penetrate through materials, meaning that anisotropy uncertainties may affect the calculated fluence. Some studies have indeed recognized that scattering angle effects are not negligible in fast reactor cores and shielding analyses [4]. Despite this, propagation of angular distribution (AD) uncertainties is not usually performed, largely due to historical lack of covariance data and the complexity of handling angular data in nuclear data processing [5]. Modern general-purpose nuclear data libraries often include comprehensive covariance matrices for reaction cross-sections, but provide little or no covariance information for scattering angles. Moreover, different evaluations of the same reaction can show different scattering anisotropy, indicating significant model uncertainties. Such inter-library

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discrepancies imply that the true uncertainty in the angular distribution can be sizable. These observations motivate a closer examination of how angular distribution covariances impact calculated neutron fluence in RPV.

In this work, we quantify the effect of Fe-56 elastic-scattering angular-distribution uncertainties on fast neutron fluence ($E > 1$ MeV) calculations relevant to PWR RPVs. We focus on Fe-56 for being the main isotope in RPV steel. It is examined how the magnitude and structure of these angular-distribution covariances influence uncertainty propagation in two neutron transport configurations and compares the behavior predicted by the latest JENDL and JEFF evaluations.

2. METHODOLOGY

2.1. Uncertainty Propagation Methods

Two complementary approaches were used to propagate nuclear data uncertainties: (1) the first-order second-moment (FOSM) method [6], also known as the "sandwich" method [7], and (2) the Monte Carlo Sampling (MCS) approach [8].

In the linear sensitivity method, the relative uncertainty in a response R (e.g., a fluence or reaction rate) is obtained by combining sensitivity coefficients with covariance data in a matrix multiplication. If $S_i = \partial R / \partial x_i$ is the sensitivity of R to the nuclear data parameter x_i, and $\mathrm{Cov}(x_i, x_j)$ is the covariance between parameters i and j, the variance of R can be approximated by the following expression:

$$\sigma_R^2 \approx \sum_i \sum_j S_i \, \mathrm{Cov}(x_i, x_j) \, S_j, \tag{1}$$

The method assumes small uncertainties and linear response, requiring the sensitivity vector **S** and the covariance matrix. Regarding frequently used MC codes, MCNP6.3 cannot compute angular-distribution sensitivities for fixed-source problems, while SERPENT2.2 can, using Generalized Perturbation Theory [9], though with notable memory demands [10]. SERPENT is therefore used here.

On the other hand, the MCS approach propagates uncertainties by sampling nuclear data from their evaluated covariances and repeating the transport calculation for each realization. The resulting spread of the score of interest quantifies the propagated uncertainty. Existing tools like SANDY [11] and FRENDY [12], support sampling from common data formats but at the time of this work, neither provides sampling for angular distributions. Therefore, a custom Python script was developed to perturb the Legendre coefficients in the ENDF-6 [13] File 4 and applied to Fe-56 elastic scattering using covariance data from File 34. The code is available upon request.

2.2. Representation of Angular Distributions and Covariances

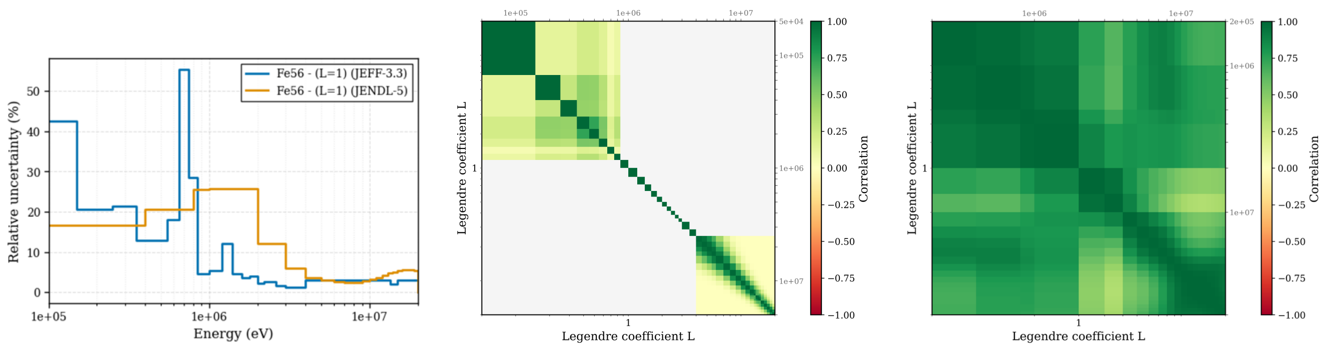
To clarify the data used in this work, it is useful to recall how elastic-scattering angular distributions are represented in the ENDF-6. Angular distribution may be specified as isotropic, with tabulated values depending on incident energy or as a polynomial expansion in Legendre coefficients. Regardless of the formatting, they all describe the same probability density function (PDF) that can be written as a Legendre series:

$$f(\mu, E) = \sum_{\ell=0}^{L_{\max}} \frac{2\ell + 1}{2} \, a_\ell(E) \, P_\ell(\mu), \tag{2}$$

where $\mu = \cos\theta$ is the cosine of the scattering angle, $P_\ell(\mu)$ are Legendre polynomials, and $a_\ell(E)$ are the energy-dependent Legendre coefficients describing the angular shape. This expression is general and may be defined in either the laboratory or center-of-mass frame, as indicated by the evaluation.

The Legendre coefficients $a_\ell(E)$ are provided in ENDF-6 File 4. Following the normalization used in Eq. (2), a valid angular PDF requires $a_0(E) = 1$; consequently, ENDF-6 omits a_0 and only lists coefficients for $\ell \geq 1$. Because $\int_{-1}^{1} P_\ell(\mu)\,d\mu = 0$ for all $\ell \geq 1$, perturbing these higher-order coefficients preserves the unit integral of the PDF. Thus, a_0 remains fixed and does not influence the sensitivities of the other a_ℓ terms. The first anisotropic contribution is the $\ell = 1$ term, $a_1(E)P_1(\mu)$, where $a_1(E)$ is numerically equivalent to the mean scattering cosine $\bar{\mu}(E)$. For use in Monte Carlo transport codes, processing tools such as NJOY [14] convert these Legendre expansions into tabulated probability distributions in the ACE format. The associated uncertainty information is found in File 34 (MF=34), which contains the variances and energy-dependent correlations for $a_\ell(E)$. These covariances must be interpreted in the same reference frame (laboratory or center-of-mass) as the MF=4 data to ensure consistent uncertainty propagation.

For transport calculations, we used JEFF-4.0 and JENDL-5 [15]. The latter includes its own covariance data from 2021 evaluation, whereas JEFF-4.0 does not provide these covariances; therefore, for the JEFF case we combined JEFF-4.0 central values with the JEFF-3.3[16] covariance set, which was inherited from JEFF-3.1 (2005). Figure 1 compares the Fe-56 elastic-scattering angular-distribution uncertainties and covariances from JEFF-3.3 and JENDL-5. Panel (a) shows the one-standard-deviation uncertainties of the first-order Legendre coefficient $a_1(E)$. Above 4 MeV, both libraries predict similar uncertainties (about 3–4%), whereas between 1 and 4 MeV, JENDL-5 shows significantly larger values (up to ∼20%) compared with JEFF (below 10%). At lower energies, JEFF shows stronger energy-dependent fluctuations, while JENDL-5 remains smoother. Panels (b)–(c) present the energy–energy covariance matrices for $a_1(E)$, revealing that JENDL-5 features stronger cross-energy correlations than JEFF. JENDL-5 provides covariances only for first-order coefficients, whereas JEFF includes up to sixth order; however, inter-order correlations in JEFF were found to be very few.



(a) 1σ uncertainty of $a_1(E)$ *(b) Correlation matrix (JEFF-3.3)* *(c) Correlation matrix (JENDL-5)*

Figure 1. Fe-56 elastic-scattering angular-distribution (AD) uncertainties and correlations. (a) One-standard-deviation uncertainty of the first-order Legendre coefficient $a_1(E)$ from JEFF and JENDL. (b)–(c) Correlation matrices of the Legendre coefficients $a_\ell(E)$ across incident energies for JEFF-3.3 and JENDL-5, respectively.

2.3. Description of Models and Simulations

To evaluate the impact of angular distribution uncertainties, we considered two neutron transport models: a simplified spherical attenuation model and an established benchmark experiment from the SINBAD database [17].

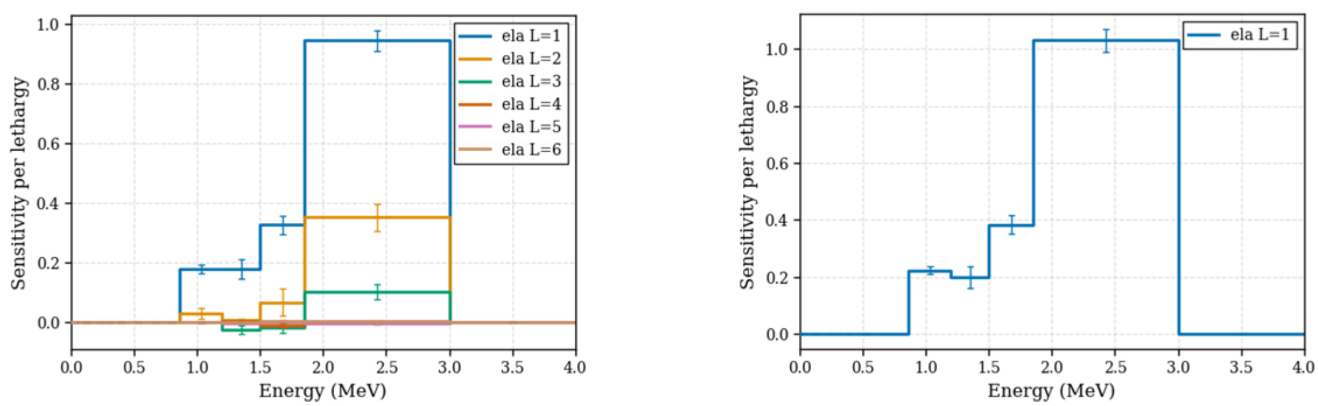
The first model is a spherical multilayer configuration inspired by the PWR core–vessel path, following the specification given in Ref. [18]. A point isotropic source at the center emits fast neutrons in the 1–3 MeV range. Concentric shells of steel and water approximate a one-dimensional core-to-RPV trajectory; the stack places a total of 34.5 cm of steel in front of the detector. A thin spherical tally region is placed just outside the final steel layer to record the 1–3 MeV fast-neutron fluence. This toy model configuration provides a clean deep-penetration case: neutrons must undergo multiple (mostly forward) scatterings to reach the detector, with significant contribution from Fe-56 elastic scattering. Simulations used the Monte Carlo transport codes SERPENT2.2 for sensitivity analysis and MCNP6.3 for sampling with 1024 realizations.

The second model is the ASPIS IRON-88 benchmark [19], from the SINBAD compilation, conducted at the Winfrith NESTOR reactor using a fission plate source and an array of 13 mild steel plates with interleaved activation foils to measure the neutron distribution. We analyzed the activation of aluminum foils via the ^{27}Al(n,α)^{24}Na reaction (threshold above 6 MeV). Foils at positions A3–A7, corresponding to increasing iron thickness ($\sim$25 cm steel thickness in A7). This benchmark represents an energy-dependent attenuation problem dominated by fast neutrons and allows comparison of calculated and experimental uncertainties (5% at 1-σ reported). Uncertainty propagation was performed with MCNP6.3 using 512 samples and the Fast TMC method, following the approach in Ref. [18]. Although the primary focus is on the impact of Fe-56 elastic-scattering angular-distribution uncertainties, cross-section uncertainties for all Fe-56 reactions with available covariance data in each evaluated library were also propagated under the same conditions to provide a consistent basis for comparison.

3. RESULTS

3.1. Spherical Attenuation Model

Figure 2 shows the sensitivity of the 1–3 MeV detector flux to the Legendre coefficients of Fe-56 elastic scattering. The first order shows the largest (and positive) sensitivity, consistent across libraries. Sensitivities remain positive for all orders, indicating that increasing the coefficients increases the response. This reflects the physical role of each term: a_1 enhances forward scattering, while a_2 sharpens the forward/backward peaks relative to lateral scattering. Orders above $\ell = 3$ are negligible.



(a) JEFF-4.0

(b) JENDL-5

Figure 2. Sensitivity of neutron flux between 1-3 MeV to Legendre coefficients for (a) JEFF-4.0 and (b) JENDL-5.

Using the sensitivity profiles, the AD and cross-section uncertainties of Fe-56 were propagated via FOSM, yielding the results summarized in Table I. In this step the sensitivities were treated as fixed, meaning that the statistical uncertainty of the sensitivity coefficients (shown as error bars in Fig. 2) was not propagated. JEFF-4.0 predicts negligible impact from AD uncertainties, with flux effects below 1%, while JENDL-5 shows significantly larger values, around 10%, consistent with its higher a_1 uncertainties in the energy range.

Monte Carlo sampling was performed perturbing cross sections (XS) and angular distributions (AD). For JEFF, additional tests confirmed no interaction between the effects due to AD and cross sections. The results, summarized in Table I, show close agreement with the FOSM estimates. JENDL-5 yields substantially higher uncertainties for both cross sections and AD, with AD contributing a non-negligible fraction of the total. In contrast, JEFF-4.0 uncertainties are smaller and essentially unaffected by AD variations.

Table I. Uncertainty decomposition for spherical model (FOSM/MCS). MCS values validated via bootstrap (95% CI with less than 5% error).

	JEFF-4.0				JENDL-5			
	FOSM		MCS		FOSM		MCS	
Metric	**AD**	**XS**	**AD**	**XS**	**AD**	**XS**	**AD**	**XS**
Propagated uncertainty (%)	0.77	11.19	0.75	10.94	10.08	23.00	11.47	22.18
Total ND unc. (%)	11.22		10.97		25.11		24.97	

3.2. ASPIS Iron Shielding Benchmark

Only MCS results are reported for ASPIS, as memory issues prevented FOSM sensitivity calculations in SERPENT. Figure 3 shows the calculated-to-experimental (C/E) ratios for the nominal cases with negligible statistical uncertainties and the averaged results from the sampled perturbations (error bars represent 1σ nuclear-data uncertainty). For reference, ENDF/B-VIII.1 nominal results are also shown, without uncertainty propagation, as the evaluation does not include AD covariances for Fe-56; the last available ENDF/B covariance set (from version VII.1) is in fact the same as that used in JEFF. Both ENDF/B-VIII.1 and JEFF-4.0 increasingly overpredict with depth, whereas JENDL-5 shows the best nominal agreement, maintaining an approximately constant C/E ratio with thickness.

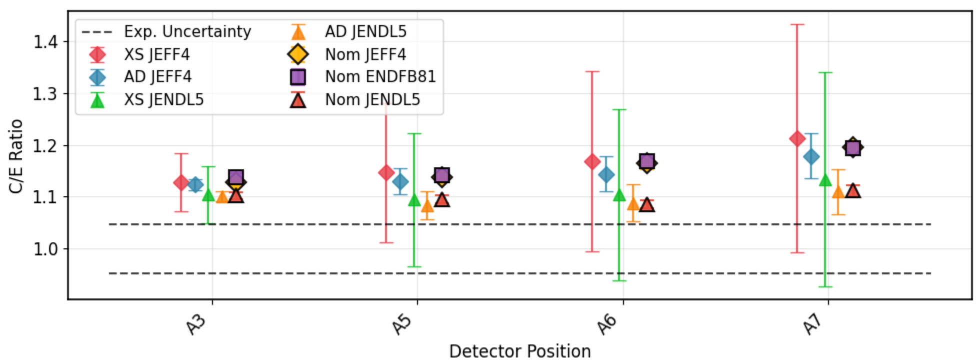


Figure 3. ASPIS C/E ratios for JEFF-4.0, JENDL-5, and ENDF/B-VIII.1. Symbols show nominal results; error bars represent 1σ nuclear-data uncertainties from MCS of cross sections (XS) and angular distributions (AD).

Figure 4 presents the propagated uncertainties for each detector. JEFF-4.0 and JENDL-5 yield comparable total uncertainties. The angular-distribution component is small, contributing about 4% of the total nuclear-data variance. This modest effect stems from the high neutron energies (E>6 MeV), where AD covariances on $\overline{\mu}$ are small.

4. DISCUSSION

To evaluate whether the AD covariance data captures realistic uncertainties, we compared the evaluated libraries and EXFOR [20] experimental measurements, treating them as representative of the range of outcomes that the covariance model is expected to cover. To interpret the different uncertainty quantification (UQ) results in the 1–3 MeV range, we plotted each library's nominal curve together with the most extreme positive/negative perturbations from 1024 samples generated between 0.5 and 7 MeV.

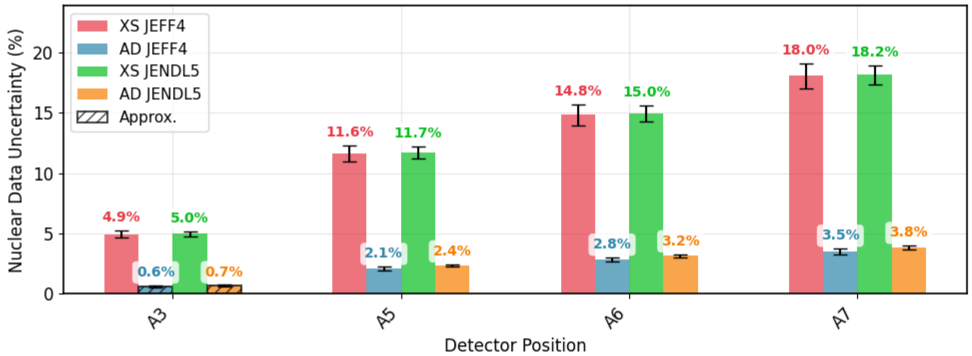


Figure 4. Propagated ND uncertainties for ASPIS Al-27 detectors from Monte Carlo sampling of cross sections (XS) and angular distributions (AD) using JEFF-4.0 and JENDL-5. Error bars indicate 95% confidence intervals. Hatched bars denote approximate ND uncertainty when Monte Carlo statistics are insufficient to reliably separate nuclear-data effects from total variance.

Figure 5 shows representative cases at 1.3 and 2.85 MeV for JEFF-4.0 and JENDL-5.

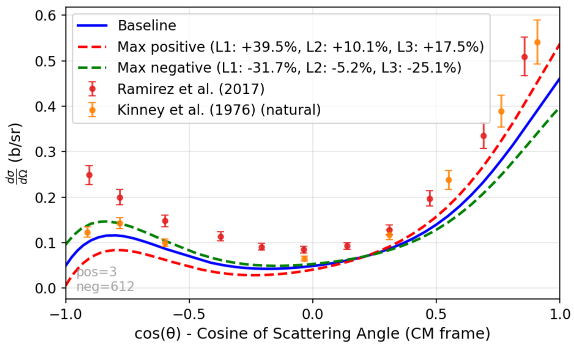
The plotted quantity, $\frac{d\sigma}{d\Omega}$, corresponds to the differential cross section as a function of the scattering angle. In addition to exploring the effect of AD uncertainties, we also assessed how uncertainties in the elastic cross sections influence these results. For JEFF-4.0, the additional variability introduced by cross-section uncertainties remains insufficient to bridge the gap with experimental data. In contrast, for JENDL-5, when both AD and cross-section uncertainties are included, the resulting uncertainty envelope provides a closer and more consistent agreement with the measurements.

Figure 5b shows that this improved agreement in JENDL-5 comes at the cost of large first-order uncertainties. This likely stems from missing covariance information in higher Legendre orders: if second- or third-order terms carried uncertainties, the same variability could be achieved with smaller L_1 spreads. In contrast, JEFF-4.0 fails to reproduce the experimental behavior even when several orders are perturbed. At 2.85 MeV (Figures 5c and 5d), JENDL-5 maintains a reasonable match to the data, particularly in the forward-peaked region, which is relevant for attenuation-dominated PWR problems. Above 3 MeV, both libraries align closely with measurements, consistent with the smaller uncertainties observed in Figure 1a.

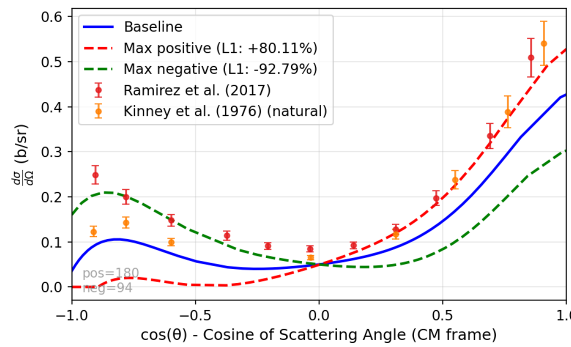
5. CONCLUSION

This study quantified how Fe-56 elastic–scattering angular-distribution covariances affect fast-neutron fluence, by combining a deep-penetration spherical model ($E \in [1\text{–}3]$ MeV) with the ASPIS Iron-88 benchmark ($E > 6$ MeV), using FOSM and MCS with JEFF-4.0 (with JEFF-3.3 AD covariance) and JENDL-5 libraries. In the sphere, JENDL-5 AD covariances yield non-negligible fluence uncertainties, whereas JEFF-4.0 remains negligible $< 1\%$. FOSM and MCS results agree, indicating near-linear behavior. In ASPIS, total nuclear-data uncertainty is similar for both libraries and largely driven by cross sections, with small AD contributions.

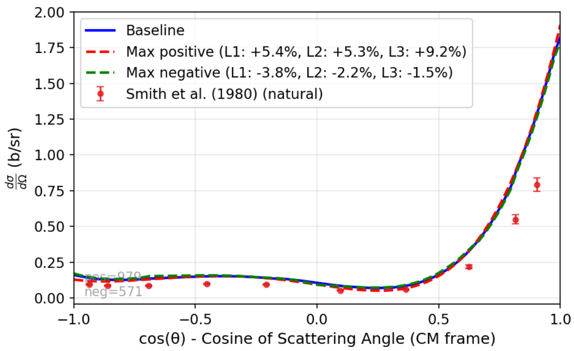
Consistency checks against EXFOR data clarify the energy dependence: above $\sim$3 MeV, evaluations align with measurements and uncertainties are small, explaining the similar ASPIS results; below 3 MeV, JENDL-5 generally covers the experimental points more effectively than JEFF libraries. Overall, the results highlight room for improvement in Fe-56 AD covariances: JEFF relies on an older evaluation (2005) that fails to cover the experimental data, whereas JENDL-5 achieves better consistency but through very large first-order uncertainties, likely compensating for missing higher-order covariances. JENDL-5 therefore provides a more realistic representation of the underlying uncertainties. Future work will



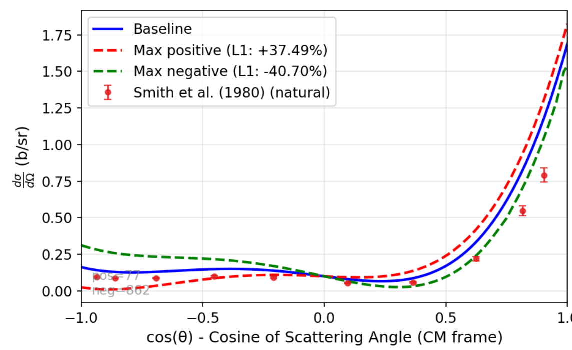
(a) 1.3 MeV — JEFF-4.0



(b) 1.3 MeV — JENDL-5



(c) 2.85 MeV — JEFF-4.0



(d) 2.85 MeV — JENDL-5

Figure 5. Fe-56 elastic angular distributions compared with EXFOR data. Blue: nominal evaluation; red/green: most extreme positive/negative perturbations among 1024 samples. Panels (a,b) show 1.3 MeV; panels (c,d) show 2.85 MeV. The cross section used for normalization is calculated at the nominal energy without folding for experimental energy resolution.

focus on generating TALYS-based ADs with alternative optical-model options and covariance data and extending the study to additional ASPIS reaction rates with lower energy thresholds and other PWR-relevant benchmarks that will help consolidate these conclusions.

REFERENCES

[1] US. Nuclear Regulatory Commission. Office of Nuclear Regulatory Research. "Regulatory Guide 1.190 Calculational and Dosimetry Methods for determining Pressure Vessel Neutron Fluence." URL http://catalog.gpo.gov/F/?func=direct&doc_number=000630223&format=999. Medium: Language material, Electronic resource, Computer, Online resource.

[2] A. Vasiliev et al. "Exploring Stochastic Sampling in Nuclear Data Uncertainties Assessment for Reactor Physics Applications and Validation Studies." **volume 9**(12), p. 1039. URL https://www.mdpi.com/1996-1073/9/12/1039. Number: 12 Publisher: Multidisciplinary Digital Publishing Institute.

[3] D. Rochman et al. "Efficient Use of Monte Carlo: Uncertainty Propagation." **volume 177**(3), pp. 337–349. URL https://doi.org/10.13182/NSE13-32. Publisher: Taylor & Francis _eprint: https://doi.org/10.13182/NSE13-32.

[4] A. Vasiliev et al. "On the importance of the neutron scattering angular distributions for the LWR fast neutron dosimetry." URL https://inis.iaea.org/records/1watf-87x71.

[5] S. Maruyama et al. "Quantifying uncertainty induced by scattering angle distribution using maximum entropy method." **volume 205**, p. 110591. URL https://www.sciencedirect.com/science/article/pii/S0306454924002548.

[6] A. Haldar et al. *Probability, Reliability and Statistical Methods in Engineering Design*, volume 23. John Wiley & Sons, Inc. URL https://linkinghub.elsevier.com/retrieve/pii/S0167473000000254. ISSN: 01674730 Issue: 1 Journal Abbreviation: Structural Safety.

[7] W. Wieselquist et al. "SCALE 6.3.1 User Manual." URL https://www.osti.gov/biblio/1959594.

[8] A. J. Koning et al. "Towards sustainable nuclear energy: Putting nuclear physics to work." **volume 35**(11), pp. 2024–2030. URL https://www.sciencedirect.com/science/article/pii/S0306454908001813.

[9] M. Aufiero et al. "A collision history-based approach to sensitivity/perturbation calculations in the continuous energy Monte Carlo code SERPENT." **volume 85**, pp. 245–258. URL https://www.sciencedirect.com/science/article/pii/S0306454915002650.

[10] P. Griveaux et al. "Computation of sensitivity coefficients in fixed source simulations with SERPENT2." **volume 200**, p. 114191. URL https://www.sciencedirect.com/science/article/pii/S0920379624000450.

[11] L. Fiorito et al. "Nuclear data uncertainty propagation to integral responses using SANDY." **volume 101**, pp. 359–366. URL https://www.sciencedirect.com/science/article/pii/S0306454916305278.

[12] K. Tada et al. "Development of ACE file perturbation tool using FRENDY." **volume 60**(6), pp. 624–631. URL https://doi.org/10.1080/00223131.2022.2130463. Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/00223131.2022.2130463.

[13] D. A. Brown. "ENDF-6 Formats Manual - Data Formats and Procedures for the Evaluated Nuclear Data Files ENDF/B-VI, ENDF/B-VII and ENDF/B-VIII." URL https://www.osti.gov/biblio/2007538.

[14] R. Macfarlane et al. "The NJOY Nuclear Data Processing System, Version 2016." URL https://www.osti.gov/biblio/1338791.

[15] O. Iwamoto et al. "Japanese evaluated nuclear data library version 5: JENDL-5." **volume 60**(1), pp. 1–60. URL https://www.tandfonline.com/doi/full/10.1080/00223131.2022.2141903.

[16] A. J. M. Plompen et al. "The joint evaluated fission and fusion nuclear data library, JEFF-3.3." **volume 56**(7), p. 181. URL https://doi.org/10.1140/epja/s10050-020-00141-9.

[17] I. A. Kodeli et al. "SINBAD – Radiation shielding benchmark experiments." **volume 159**, p. 108254. URL https://linkinghub.elsevier.com/retrieve/pii/S0306454921001304.

[18] J. A. Monleon de la Lluvia et al. "Toward Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations." pp. 1–23. URL https://doi.org/10.1080/00295639.2025.2510048.

[19] M. Pescarini et al. "Preliminary Results of the BUGJEFF311.BOLIB Library Validation on the Iron-88 (Fe) Neutron Shielding Benchmark Experiment." URL https://iris.enea.it/retrieve/dd11e37c-db68-5d97-e053-d805fe0a6f04/ADPFISS-LP1-064.pdf.

[20] IAEA. "Experimental Nuclear Reaction Data." URL https://www.iaea.org/resources/databases/experimental-nuclear-reaction-data. Publisher: IAEA.